



Unit 4 · Outcome 3 · Scientific Investigation

The nature of evidence

KK06

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 06 The nature of evidence Science doesn't run on belief, authority or how loudly someone argues

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



FOUNDATIONS

When does data become evidence?

Students often use “data” and “evidence” as synonyms. They aren’t. The difference is the single most useful idea in this dot point.

- **NOTE**

D Data are the raw recorded observations and measurements — Mara’s DO readings of 8.4, 8.1, 6.2 and 5.9 mg/L at four sites. On their own, numbers in a logbook prove nothing.

- **NOTE**

E Evidence is data interpreted in relation to a specific claim. The moment Mara says “these falling DO values support my hypothesis that the drain lowers water quality,” the data has become evidence for something.

- ✓ **TIP**

✓ One-line definition to memorise: evidence is data that has been used to support or challenge a hypothesis, model or theory. The same dataset can be strong evidence for one claim and irrelevant to another — what makes it “evidence” is the claim it’s pointed at.

A good analogy: data are the fingerprints, CCTV and timestamps at a scene. They only become evidence when a prosecutor connects them to a specific accusation. The same CCTV clip might convict one suspect and exonerate another. Evidence is always evidence for or against a particular claim.

THE THREE THINGS EVIDENCE IS WEIGHED AGAINST

Hypothesis, model & theory

The dot point names three different targets that evidence can support or refute. Examiners love this because students routinely muddle them — especially the idea that “theory” means “a hunch”. It’s the opposite. Click each tier in the model below.

- **NOTE**

The big misconception. A hypothesis does not “grow up” into a theory, and a theory does not graduate into a “law” once it’s proven. They are different kinds of statement, not stages of certainty. A theory is already the strongest form of scientific explanation — a huge body of evidence resting on it — not a guess waiting for promotion.

Mara: “DO is lower downstream of the drain than upstream.” Testable in an afternoon.

A climate model, the carbon cycle, or a predator–prey graph — a representation you run to make predictions.

The enhanced greenhouse effect; evolution by natural selection. Decades of converging evidence.

THE VERDICT

Does the evidence support or refute?

Imagine a set of scales. On one pan, evidence consistent with the claim; on the other, evidence that contradicts it. The job of a scientist — and of you in the exam — is to read which way the beam tips, and how confidently. Add evidence to Mara's scales and watch the verdict change.

- **NOTE**

i Supporting evidence raises confidence; it never “proves”. Even a beam tipped hard toward “support” only means the hypothesis is well-supported so far. Science deals in confidence, not proof. The honest exam verb is “supports”, not “proves”.

WHY THE TWO PANS AREN'T EQUAL

Falsifiability & the asymmetry of proof

Here is the idea that separates an A+ answer from a B. Supporting and refuting evidence are not symmetrical. You can pile up a thousand confirmations and still not prove a hypothesis — but a single solid, reproducible contradiction can falsify it. A claim that could never be contradicted by any possible observation isn't scientific at all.

- ✓ **TIP**

✓ Falsifiable = scientific. “DO is lower downstream” is falsifiable — one careful measurement could contradict it. That's what makes it a real hypothesis you can test.

- **NOTE**

Unfalsifiable = unscientific. “The creek has a spirit that resists pollution” can't be contradicted by any measurement, so no evidence could ever bear on it. Outside science.

- **NOTE**

i But one odd result doesn't instantly kill a theory. A well-supported theory has so much converging evidence that a single anomaly usually points to experimental error or a special case first — investigated, replicated, and only then taken to revise the theory. Refutation is powerful, but it still has to be reproducible. This is why scientists chase anomalies rather than ignore them.

NOT ALL EVIDENCE WEIGHS THE SAME

Weight & quality of evidence

A scientist doesn't just count evidence — they weight it. Reproducible, peer-reviewed data from independent sources outweighs a single uncontrolled observation. The accumulated, converging weight of evidence is what builds scientific consensus — which is why “a study found” is much weaker than “decades of independent studies all find”.

EXAM CRAFT

Decode the command word

◆ COMMAND WORDS

This dot point shows up under specific verbs. Each one wants a different shape of answer. Tap a verb to see what the examiner is really asking for.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

LOSE-MARKS-HERE MOMENTS

Six exam traps

⚠ EXAM TRAPS

Every one of these is a real way students drop marks on “nature of evidence” questions. Tap to reveal the fix.

WRITE LIKE AN EXAMINER WANTS

P·E·L trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

High-scoring evidence answers follow a rhythm: make a Point, cite the Evidence, then Link it back to whether the hypothesis is supported or refuted. Write a response below — the meters light up as you hit each move. The evidence supports Mara’s hypothesis. Dissolved oxygen falls consistently from 8.4 mg/L upstream to 5.9 mg/L downstream, which is the pattern her hypothesis predicts, and because the trend repeats across three sampling days it is reproducible rather than a one-off fluctuation, making it stronger evidence. However, it raises confidence rather than proving the claim — the drain has not been isolated as the only cause, so the evidence supports but does not confirm the hypothesis.

SEE THE DIFFERENCE

Weak vs strong: a worked answer

Question (4 marks): “Distinguish between a hypothesis and a theory, using an example of each from environmental science. Explain what kind of evidence relates to each.”

• NOTE

Why it loses marks: “theory = proven hypothesis” is the classic error; theories aren’t promoted hypotheses. “Climate change” is vague (the enhanced greenhouse effect is the theory). “Evidence proves” misstates how science works.

• NOTE

Why it scores full marks: clear scope contrast (narrow vs broad) ✓ · correct example of each ✓ · the evidence distinction (one study vs converging body) ✓ · avoids “proof” language ✓.

TEST YOURSELF · 21 MARKS

Practice questions

★ PRACTICE & SELF-MARK

Attempt each in your logbook first, then reveal the marking guide and grade yourself honestly. Your score builds in the dock (bottom-right).

LOCK IT IN

Recap — the seven things to remember

✦ KEY TAKEAWAY

Data ≠ evidence Data become evidence only when interpreted in relation to a specific claim.

✦ KEY TAKEAWAY

Three different targets Hypothesis (narrow, testable), model (a representation/tool), theory (broad, well-supported explanation).

✦ KEY TAKEAWAY

Theory ≠ guess A theory is the strongest form of explanation, not a hypothesis awaiting promotion. Theories don’t become “laws”.

✦ KEY TAKEAWAY

Support, don’t prove Evidence raises or lowers confidence. Science doesn’t “prove”; use “supports”.

✦ KEY TAKEAWAY

The asymmetry No amount of confirmation proves a claim, but one reproducible contradiction can falsify it.

✦ KEY TAKEAWAY

Falsifiable = scientific If no possible observation could ever contradict a claim, it isn't science.

✦ KEY TAKEAWAY

Weigh, don't count Reproducible, controlled, independent, peer-reviewed evidence outweighs anecdote — consensus is converging weight.

✦ KEY TAKEAWAY

Provisional, not flimsy Even strong theories stay open to revision by new reproducible evidence — that's a strength, not a weakness.

• NOTE

Next in the chapter: KK07 — Analysing primary data: organising, analysing and evaluating data to find patterns and relationships, including sources of error and uncertainty. That's where Mara turns her evidence into a defensible conclusion.

SCIENTIFIC INVESTIGATION · KEY KNOWLEDGE 06

The nature of evidence

Science doesn't run on belief, authority or how loudly someone argues. It runs on evidence — and on a careful, almost lawyerly judgement about whether that evidence supports or refutes the idea on trial. This module is about how a scientist weighs evidence, and why a hypothesis, a model and a theory are three very different things on that scale.

the nature of evidence that supports or refutes a hypothesis, model or theory

• NOTE

Running investigation for this module. Meet Mara, a Year 12 student studying Tanderra Creek, an urban waterway in suburban Melbourne. A large stormwater drain empties into it. Mara's hypothesis: if urban stormwater lowers water quality, then dissolved oxygen (DO) will be lower downstream of the drain than upstream. We'll follow her evidence the whole way through — and zoom out to climate science when we need a model and a theory.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

X WEAK ANSWER

Unfalsifiable = unscientific. “The creek has a spirit that resists pollution” can’t be contradicted by any measurement, so no evidence could ever bear on it. Outside science.

✓ STRONG / MODEL ANSWER

A hypothesis is a narrow, testable prediction — e.g. “DO is lower downstream of a stormwater drain” — which a single controlled investigation can support or refute. A theory is a broad, well-substantiated explanation supported by a large converging body of evidence — e.g. the enhanced greenhouse effect. A hypothesis relates to direct primary data from one study; a theory rests on decades of independent, reproducible evidence and is the strongest form of explanation, not an unproven guess. Why it scores full marks: clear scope contrast (narrow vs broad) ✓ · correct example of each ✓ · the evidence distinction (one study vs converging body) ✓ · avoids “proof” language ✓.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.